SoftClient4ES

BENCHMARK REPORT

Extracting data out of Elasticsearch

SoftClient4ES over Arrow Flight SQL, measured against Trino's Elasticsearch connector — same host, same cluster, same 10-million-row index, back to back.

DATASET	SOFTCLIENT4ES	TRINO	ELASTICSEARCH
10,000,000	**0.3.0**	**483**	**8.18.3**
rows · 8 columns	Arrow Flight SQL sidecar	Elasticsearch connector	3-node cluster

1 Summary

This benchmark measures how efficiently each system extracts data **out of Elasticsearch** into a Python data-science client (pandas, polars, DuckDB). Both run against the same 3-node Elasticsearch cluster, the same 10-million-row index across 6 primary shards, on the same host, back to back.

> **Who this is for:** data teams pulling large result sets out of Elasticsearch into Python. It is **not** a distributed-analytics, federation, or SQL-coverage comparison — for those, Trino is the right tool and this benchmark deliberately does not exercise its strengths.

AGGREGATION PUSHDOWN

27.1 KB vs 1.39 GB

Data moved off the cluster for a `GROUP BY` returning 100 rows. SoftClient4ES compiles it into an Elasticsearch aggregation; Trino scans all 10M rows into its own engine. The one structural result here, *against Trino*: unchanged in kind at 1 and 6 shards — 25.9 KB and 27.1 KB off the cluster against 1.39 GB — grounded in the connector's documentation — every other multiple can erode; this one moves only if the connector changes.

CLIENT MEMORY FLOOR

2 GB container

Lands all 10M rows as a DataFrame inside a hard 2 GB cap. Trino's fastest client is killed there; its documented client is killed at every cap measured. Streamed instead of materialised, both engines run everywhere (§5).

CONCURRENCY IN AN 8 GB BUDGET

5 vs 2

Five simultaneous 10M-row extractions complete — 50 million rows — against two for Trino's fastest client, and none for its documented one. A 2.5× capacity ratio, not five-versus-nothing.

10M-ROW EXTRACTION

1.23–3.75× faster

11.91 s against 14.66 s for Trino's fastest client (connectorx, on 3.7× less client CPU — though it uses less memory than we do) and 44.70 s for its documented one (8.5× less CPU, 4.8× less memory). **The low end is the serious comparison** — nobody extracts 10M rows through the documented client on purpose; quote the route with the multiple.

For work Elasticsearch can compute, SoftClient4ES pushes it into the cluster and moves almost no data off it — and that result leads this report deliberately: it is its one **structural** finding *against Trino*, unchanged in kind at 1 shard and at 6 — the 100-row answer travels, the 10M rows never do — grounded in the connector's own documentation and exact by Elasticsearch's own error bounds, where every wall-clock multiple below is competitive and can erode. It is not a property no one else has: Elasticsearch's own ES|QL pushes the same aggregation down and ties us on that cell (§4). What is structural is the difference from a connector that pushes no aggregation down at all. For large extractions it runs in far less client memory, and it is faster, because it delivers Arrow columnar batches the client consumes directly where Trino's Python client materialises rows into Python objects. Trino runs throughout as a **3-node cluster allocated 1.5× our CPU and 2× our memory** — 6 CPUs and 8 GB against the sidecar's 4 and 4, and neither side's server counts against the client budgets of §5. The handicap buys fairness, not speed, and the CPU counters say how much of it is used — **which depends on the route, so this report states both**. Driven by its documented client, Trino's 55.4 s of engine CPU over a 44.70 s run average **about 1.2 of its 6 CPUs** against our 2.4 of 4,

and total system CPU (engine plus Elasticsearch) comes to **85.8 s against our 71.0 s**. Driven by connectorx — the route every multiple here is quoted against — the same cluster averages **2.8 of its 6 CPUs**, 47% of its allocation against our 60%, and total system CPU is **73.8 s against our 71.0 s: parity**. The wide system-cost gap belongs to the documented client, not to Trino. The one column that favours Trino on *every* route is Elasticsearch itself — we cost the cluster **42.2 s of CPU against its 30.4 s** on the first-measured cell (35.7 s settled — see the warm-in note in §2), which over the run is **3.5 of the cluster's 6 CPUs against 0.68** (documented route; 2.2 on connectorx). Where that cluster also serves search traffic, it is a real operational trade-off; §A carries it as a limitation. Resharding from 6 primary shards to 1 costs us **3.50×** and Trino **1.23×**, so the gap narrows from 3.75× to 1.32× — nearly all of the headline is shard parallelism, and section 7.1 attributes it on our own side alone, by measuring the same build with concurrent paging on and off. Read that as a **lead, not a moat**: sliced paging uses Elasticsearch's public `slice` API, available to any engine — Trino's connector already plans one split per shard. What no pagination strategy changes are the architectural results: the push-down (S3) and the client cost profile.

Calibration — two reference points, not alternatives. Two things in this report extract faster than either SQL engine. Neither is a candidate for the workload this benchmark is about, and section 8 says why; they are published because a reader who knows Elasticsearch will ask, and an unmeasured question reads as an avoided one.

A hand-written sliced scroll. Six processes taking one Elasticsearch slice each — one per primary shard — building the same Arrow table we deliver, extract the same 10M rows in **13.76 s against our 11.91 s**. Stripped down to merely counting the rows and discarding them, a cheaper task than either engine performs, it manages 12.46 s — still, narrowly, slower than us. Concurrent paging, new in the release measured here, closed that gap: the floor is no longer the fastest extraction in this report. It pays **8.5× our client CPU and 4.0× our client memory** to build the artifact, which puts it among the clients that cannot complete in the 2 GB container where we do — but cost is not why it is a floor rather than an alternative. **It is not written in SQL.** This report is for data engineers, analysts and scientists moving Elasticsearch data into pandas, polars, DuckDB or a BI tool, and for that reader the floor's query is not a query at all: it is a hand-maintained Elasticsearch Query DSL document — projection, filter, sort, slice and page size as JSON — where every change to the question is an edit to that JSON. It also cannot **aggregate without re-reading everything** (the `GROUP BY` of S3 costs us 0.043 s and 27.1 KB off the cluster) and cannot **join at all** — Elasticsearch has no join, so J0–J2 are not slower on this floor, they are absent. So the scope is stated rather than implied: this compares **SQL engines over Elasticsearch**, for people who work in SQL and dataframes.

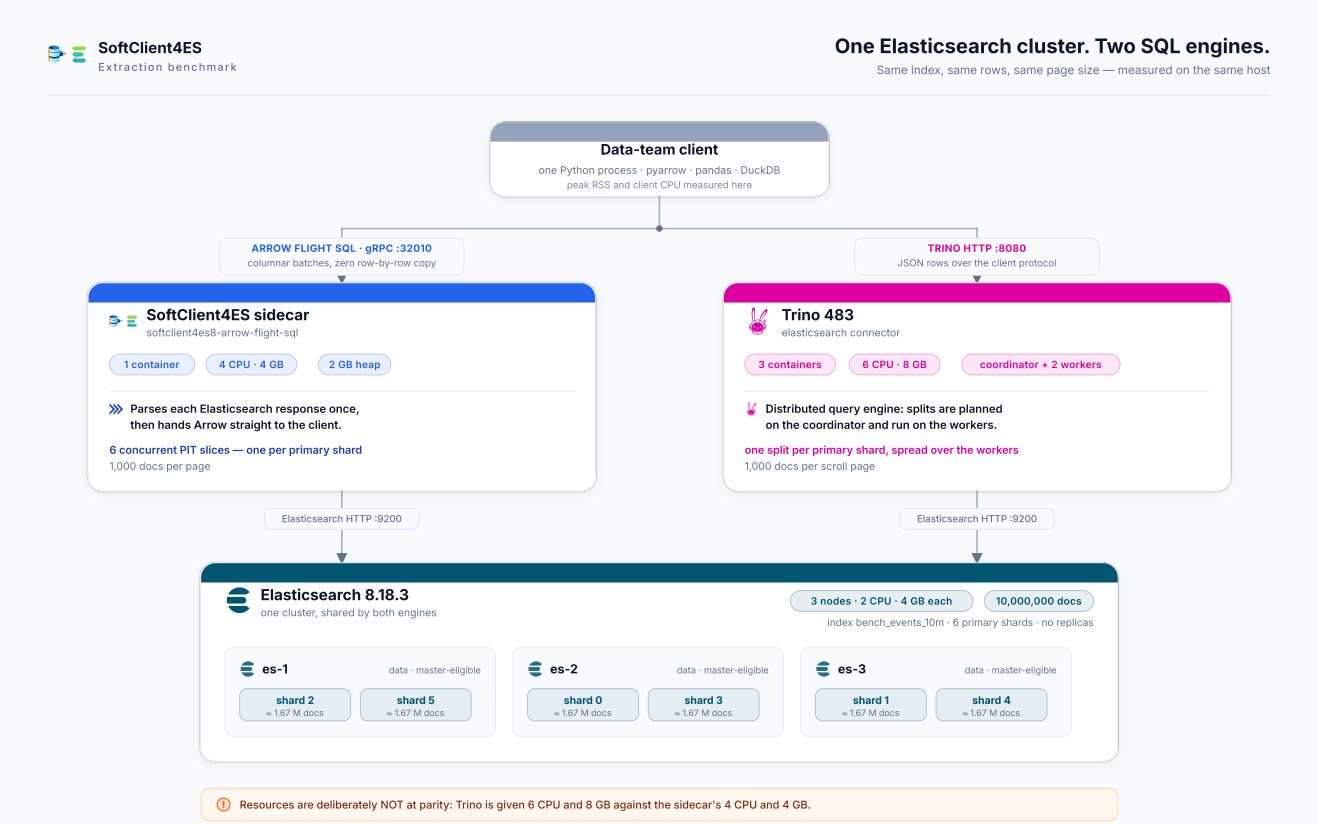
> **What this floor does and does not mean for a decision.** It is not an argument against using a SQL engine over Elasticsearch and should not be read as one. It says that *if* a workload is one fixed extraction, written once as Query DSL and maintained as JSON, with no joins and no aggregation push-down, then a script is a reasonable way to move those rows — and that is published here rather than hidden. The moment the workload is a **query language** — users writing SQL, joining indices, aggregating, landing results in pandas, polars, DuckDB or a BI tool, or running anywhere memory is bounded — the floor stops being an option at all, and the comparison that decides anything is the one between the SQL engines (sections 4 onward).

Elasticsearch's own query language. Up to and including one million rows, ES|QL extracts faster than either engine here and can hand back Arrow directly — **0.32 s against our 3.99 s** — and section 4 publishes those cells, including the ones we lose. On the pushed-down aggregation the two are level (0.034 s against our 0.043 s, inside the noise for cells this small). ES|QL cannot return more than 1,000,000 rows, which is why the ten-million-row scenarios have no ES|QL column. **Trino wins every cross-index join measured here** — by 7.0%, 12.4% and 22.3% — offers a client that uses less memory than ours, gains the most from shard parallelism on aggregate-heavy work, and provides capabilities — distributed execution, spill, connector breadth — this benchmark does not exercise (section 8).

> All figures in this report were measured on the **released** sidecar image `softnetwork/softclient4es8-arrow-flight-sql:0.3.0` (digest `sha256:9dd80b77f62e…`), Trino 483 and Elasticsearch 8.18.3. No number is published that did not come out of a run of the harness.

2 Environment

SoftClient4ES sidecar	`softclient4es8-arrow-flight-sql:0.3.0` — Arrow Flight SQL, gRPC :32010
Trino	483 (official image), Elasticsearch connector
Elasticsearch	8.18.3 — a **3-node cluster**, every node data + master-eligible, quorum of 2
Index	`bench_events_10m` — 10,000,000 documents, flat mapping, **6 primary shards**, 0 replicas, force-merged (974 MB)
Host	Apple M4 Max, macOS (Darwin arm64), 16 logical cores; Docker Desktop VM 13 CPU / 32 GB allocated (reports 31.3 GiB)
Resources, and the handicap they encode	**One** SoftClient4ES sidecar: 4 CPU / 4 GB. **Trino: a 3-node cluster totalling 6 CPU / 8 GB** — 1.5× the CPU and 2× the memory, in every scenario. Elasticsearch: 3 nodes × 2 CPU / 4 GB (heap 2 GB each)
Sensitivity topology (§7)	`bench_events_10m_s1` — same corpus, same seed, **1 primary shard**. The engines are unchanged: only the index topology differs; only one 10M index is open at a time
ES|QL	Elasticsearch's own query language, measured as a third stack wherever it can run. Its cells were taken with `esql.query.result_truncation_max_size` raised from its 10,000 default to 1,000,000 — the product maximum; every ES|QL run records the effective value.
Runs	≥2 warm-ups, then ≥5 measured runs per cell; each run in a fresh client process. Every wall-clock cell in §4 and §7 prints the median with its **min–max** in brackets; §6's joins publish theirs in a companion table, with all 25 cross-pairings. The §5 sweeps are **one run per cell** by design — their outcome is binary, completes or killed — so they carry no dispersion; the repeated cells bound run-to-run noise at ≤6%.



What was measured. Both engines read the same 3-node Elasticsearch cluster and the same 6-shard, 10-million-row index, on the same host. Shard placement is the cluster's actual allocation, not an idealised one. Resources are deliberately not at parity: Trino is given 6 CPU and 8 GB against the sidecar's 4 and 4.

Client libraries

Identical versions on both sides wherever a library is shared.

LIBRARY	VERSION	USED BY
CPython	3.12.12 (arm64)	all
pyarrow	25.0.0	SoftClient4ES (Arrow)
adbc-driver-flightsql	1.12.0	SoftClient4ES (Flight SQL)
adbc-driver-manager	1.12.0	ADBC drivers
pandas	3.0.5 (numpy 2.5.1)	both
polars	1.43.2	both (polars variant)
duckdb	1.5.5	both (S2)
trino	0.338.0	Trino (documented client)
sqlalchemy	2.0.51	Trino (`pandas.read_sql` / `pl.read_database`)
connectorx	0.4.5	Trino (fast columnar client)
ADBC Trino driver	0.5.1	Trino (ADBC client)

Measurement provenance. The **6-shard** extraction matrix — floors, S1, S1m, S1r, S2, S3, S4, both ES|QL wire formats, Trino's connectorx and ADBC routes, every S1r destination, the tuned-Trino arm, the hostname-dial arm and the drift controls — is **one session,** `results/20260821T041841-v030-prewarm` : 180 measured runs, one gate, one host state, and — new to this session — **the corpus warmed to a verified steady state before the first timed block** (`warm-in.json`). Four groups cannot share it because each rebuilds its own container topology or index, so each is its own session, run the same morning on the same host and image: S5, S6, the joins, and section 7's 1-shard arm — as is the paging A/B of §7.1. One session is quarantined whole with a README: a prior attempt whose Trino leg died because macOS put the host to sleep mid-block; every phase now runs under `caffeinate` . Every table names the session it draws on; no table mixes two. *The warm-in — this report's one moving measurement, published against ourselves:* the session opens by extracting the full corpus, untimed, until Elasticsearch's own CPU per pass stabilises (4 passes, 45.4 → 36.7 → 35.9 → 36.4 s), so no timed block starts cold and a cold page cache is excluded by construction. Even so, the first timed S1 block measures **11.91 s at 42.2 s of Elasticsearch CPU**, and the same cell re-measured at the **end** of the session — for both engines — runs **10.04 s at 35.7 s**, while **Trino moves 0.4%** (44.70 → 44.52 s). The extra cost is server-side (the sidecar also declines, 28.8 → 24.3 s, consistent with JVM warm-up; client CPU is flat), it decays over repeated identical reads, and it re-arms: the scroll floors, run between the warm-in and the first block, pushed the sliced path's Elasticsearch cost back from ~36 to 42 s. The mechanism is **not established**; what is established is that it only affects our sliced path, on a build and cluster otherwise metronome-stable. **This report publishes the first-measured block (11.91 s) as the headline** — the state a user meets on a warm cluster — and prints the end-of-session 10.04 s as the favourable bound it deliberately does not use. Cells measured later in the block sequence (S1r, S2) sit in the settled state, which is why S1r's 9.97 s is *faster* than S1's 11.91 s for strictly more work — the order of measurement, not the workload, separates them. *Host load, and the throttling counters:* the 1-minute load average sampled with every run had a median of 5.5 and a maximum of 13.2 against 16 logical cores — 2.8 cores uncommitted at the worst moment; the harness refuses to measure above a configured ceiling. The VM's 13 CPUs are oversubscribed by the declared limits (16) with asymmetric headroom (ours 10 of 13, Trino's 12 of 13), so this session publishes what the load average cannot see: the cgroup throttling counters, per run, for every stack — **zero throttled periods for either engine and for Elasticsearch on every 10M-row cell**; the one non-zero reading anywhere is Trino's 0.8 s during the 4.4 s S1m burst. Neither engine's limit ever bound a headline measurement. *Host memory pressure:* macOS reported pressure level 1 (normal) on all 180 runs, with 16.7 GB available at the tightest moment. Swap sat flat to the megabyte at 3,810 MB across the overnight matrix, and equally flat near 9.0 GB during the morning Arrow-floor addendum (8,922 MB at the minimum and 9,066 MB at the maximum) — memory swapped out before each block, not paging during it.

Metrics. *Wall* is end-to-end time including connection. *Client CPU* is `time.process_time()` in the client process. *Peak client memory* is the process's peak physical footprint (`ri_lifetime_max_phys_footprint`), which is immune to the macOS memory compressor. One caution when reading across sections: §5's clients run *inside* Linux containers, where the instrument is `ru_maxrss` — within every scenario both engines share one instrument, so each comparison stands, but memory figures from §4 and §5 are not on one scale. *Engine CPU* and *Elasticsearch CPU* are exact cumulative counters read from each container's cgroup (`cpu.stat` / `usage_usec`), not sampled. *ES wire* is the bytes that left the Elasticsearch cluster, computed as the sum of the three nodes' `eth0` transmit counters **minus** the inter-node transport traffic they report, so replication is not counted as data delivered to a client. Historically, measured from container network counters — sampled at the Elasticsearch container itself, in every table of this report, so the figure never depends on how many containers the engine is made of. Every run asserts the exact expected row count before its timing is recorded; a run that returns the wrong number of rows is discarded, not reported.

3 Scenarios at a glance

ID	QUESTION IT ANSWERS	OUTCOME
S0	What does a single-process Python scroll client cost as a reference floor?	37.0 s to read 10M rows
S0p	And a *sliced* scroll — 6 slices, one per shard, building the same Arrow table?	**13.76 s**, for 8.5× SoftClient4ES's client CPU on S1
S1	Extract 10M rows into a client-side columnar table	**1.23× faster** than Trino's fastest client, **3.75×** than its documented one; 3.7× and 8.5× less client CPU by route
S1m	Extract **1,000,000** rows — the only scale all three stacks reach	**ES|QL 0.32 s**, SoftClient4ES 3.99 s, Trino 4.42 s
S1r	Extract 10M rows into a **pandas / polars DataFrame** (what an analyst builds)	**1.3–1.6× faster** to pandas and **1.2–1.4×** to polars than Trino's fastest route; **4.7–5.6×** and **4.3–5.1×** its documented one (band = position in the block sequence, §4)
S2	Extract 10M rows and compute an aggregate in DuckDB	**4.3–5.1× faster**, 8.0× less memory
S3	`GROUP BY` returning 100 rows	**0.043 s vs 5.50 s**, and moves **27.1 KB against 1.39 GB** off the cluster
S4	Fetch 100 rows (`LIMIT 100`)	**Parity** 37 ms vs 65 ms; ES|QL takes it at 6 ms
S5	Does the extraction fit in a constrained-memory container?	Fits **2 GB**; Trino's documented client never fits (OOM at 8 GB), its fastest client 3 GB
S6	How many concurrent extractions fit in an 8 GB budget?	SoftClient4ES **5** (50M rows); Trino **2** with its fastest client, **0** with its documented one
J0–J2	Cross-index JOIN landed as a DataFrame	**Trino faster on all three** — by 7.0%, 12.4% and 22.3%; SoftClient4ES does 5–23× less client work
§7	How much of the gap is shard parallelism? (6 shards vs 1)	6→1 shard costs us **3.50×**, Trino **1.23×**; the gap narrows from **3.75× to 1.32×**; peak client memory and pushdown unchanged, client CPU is not

All scenarios run against one 10-million-row index (`bench_events_10m`); the JOIN scenarios also use a 1-million-row index (`bench_1m`).

4 Extraction scenarios

S0 The reference floors

A plain Python client that scrolls Elasticsearch and parses each JSON hit. Two shapes: one process with one point-in-time reader, and six processes each taking one Elasticsearch slice — one per primary shard. Each shape is measured twice: counting rows and discarding them is a *cheaper task* than producing a columnar table, so both also run **building the real deliverable** — one Arrow table, 10M × 8, the same artifact S1 produces.

FLOOR (6 SHARDS)	WALL	CLIENT CPU	ELASTICSEARCH CPU	PEAK CLIENT MEMORY
S0 — one reader, count and discard	37.00 s [36.76–37.14]	14.49 s	45.5 s	27 MB
S0 — one reader, **building Arrow**	40.07 s [39.98–40.41]	17.54 s	46.7 s	3,455 MB
S0p — 6 slices, count and discard	12.46 s [12.21–12.94]	15.97 s	34.6 s	165 MB
S0p — 6 slices, building Arrow	13.76 s [13.42–14.25]	24.13 s	33.7 s	3,669 MB
SoftClient4ES S1, for comparison	**11.91 s [11.73–12.01]**	**2.83 s**	42.2 + 28.8 s sidecar	**921 MB**
Trino S1 (connectorx), for comparison	14.66 s [14.51–14.71]	10.52 s	32.9 + 40.9 s Trino	617 MB

The sliced floor does not win, and the margin is honest about being thin. Building the same artifact we deliver, it takes **13.76 s against our 11.91 s** — we are **1.16× faster**, and 1.16× is the number this report uses. (Against our settled-state block the same floor would read 1.37×; that is the favourable bound §2 discloses and deliberately does not publish as the result.) Even the count-and-discard arm — which produces nothing, and is therefore not a comparison at all — lands at 12.46 s, still behind us. §7.1 shows why: with concurrent paging disabled we take 35.34 s, and the floor beats us comfortably.

Note the slice count, because it decides the number. The floor is measured at **6 slices on a 6-shard index**. An earlier run of this session used 5 — a literal left over from a previous topology — which hands one slice two shards and makes the wall clock the slowest slice's: it measured 19.2 s instead of 12.46 s, a **31% handicap in our own favour**. Those records are quarantined rather than deleted. A floor measured at the wrong parallelism flatters whatever it is compared against, and this one is compared against us.

What separates it from an engine — measured, not asserted. Producing the artifact costs it **8.5× our client CPU** (24.13 s against 2.83 s) and **4.0× our client memory** (3,669 MB against 921 MB), so it joins the clients **OOM-killed in the 2 GB container** where SoftClient4ES completes (S5). And it answers exactly one query shape: the `GROUP BY` of S3 costs us 0.043 s and costs it a full re-scan, because a script has nothing to push down with. That is the scope boundary this report states on page 1 rather than leaving a reader to find.

S1 Extract 10M rows into a columnar client table

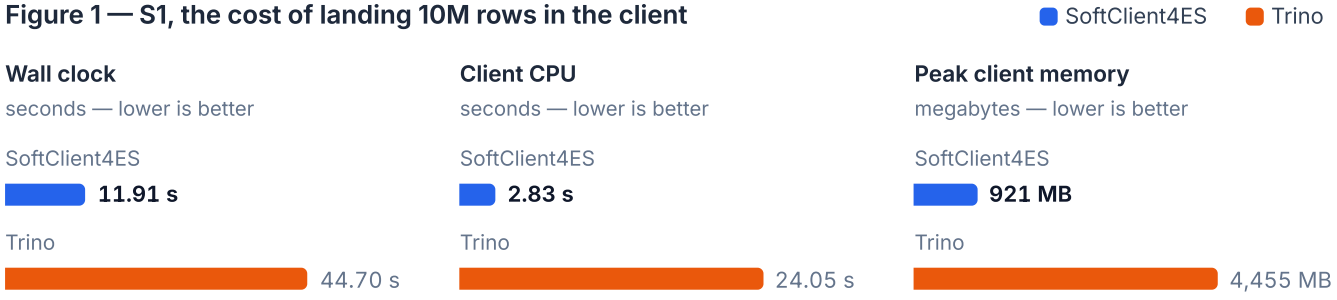
The full 10-million-row result set materialised in the client. SoftClient4ES fetches an Arrow table; Trino's documented client fetches rows.

```
# SoftClient4ES — Arrow Flight SQL (adbc_driver_flightsql)
cur.execute("SELECT id, event_ts, amount, qty, status, country, category, name "
            "FROM bench_events_10m")
table = cur.fetch_arrow_table()          # Arrow columnar batches

# Trino — documented client (trino.dbapi)
cur.execute(same_sql)
rows = cur.fetchall()                    # list of Python tuples
```

METRIC	SOFTCLIENT4ES	TRINO	RATIO
Wall median	**11.91 s** [11.73–12.01]	44.70 s [44.46–44.97]	**3.75× faster**
Client CPU	**2.83 s**	24.05 s	8.5× less
Peak client memory	**921 MB**	4,455 MB	4.8× less
Engine CPU	**28.8 s**	55.4 s	1.9× less
Elasticsearch CPU	42.2 s	**30.4 s**	1.39× *more*
ES wire	2.54 GB	2.92 GB	—
Rows / columns	10,000,000 / 8	10,000,000 / 8	✓

Figure 1 — S1, the cost of landing 10M rows in the client



Each panel has its own scale and unit; bars are comparable within a panel only. Medians of ≥5 runs after 2 warm-ups.

Outcome. SoftClient4ES is faster and far lighter. The reason is client representation: the client consumes Arrow batches without ever building 10 million Python objects, which is what dominates Trino's client CPU and memory.

The upstream wire gap is systematic and unexplained. For the same 8 columns at the same page size, Trino reads **292 bytes per row** off the cluster where we read **254 bytes** — reproduced at S1m (308 against 253 bytes), so it is behaviour, not noise. Candidates (`_source` filtering, per-hit metadata, the scroll-versus-PIT envelope) have not been separated; a capture of each engine's `_search` requests is queued, and until then this report notes the gap rather than explaining it.

The downstream leg, measured. The bytes each engine sends its client were recorded on every run, per container: the sidecar's egress is **0.71 GB** of Arrow batches for the 10M rows, while **Trino's coordinator sends 0.25 GB** to the client — its HTTP responses are compressed — the other 0.72 GB of its stack's egress being internal worker-coordinator exchange. (connectorx declines the compression: the same coordinator sends it 0.85 GB — **more** than our 0.71 GB, so the ordering inverts with the client.) Read it plainly: **on the documented route Trino ships fewer bytes to the client than we do, and on connectorx it ships more** — which is the point: the byte count does not explain the gap on either route. The client-side gap is not the byte count but what the client must do with the bytes — 24.05 s of CPU turning JSON pages into Python objects against 2.83 s consuming Arrow. This also bounds the client-boundary objection (the client runs outside the Docker VM): the leg crossing that boundary is *smaller* for Trino, and the in-container scenarios (§5), where no boundary exists, reproduce the same ordering.

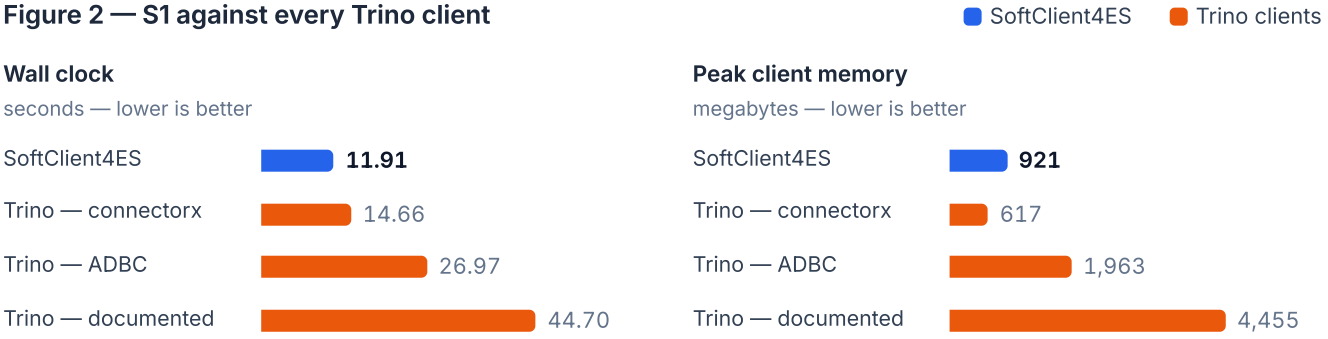
Fairness — measured against Trino's fastest clients

Trino's documented client is not its only option, and it is its slowest. **connectorx** (a Rust engine that parses Trino's result pages straight into columnar buffers) and the **ADBC Trino driver** both return an Arrow table. Measured on the same query:

ROUTE	WALL	CLIENT CPU	PEAK MEMORY
SoftClient4ES — Arrow Flight SQL	**11.91 s** [11.73–12.01]	**2.83 s**	921 MB
Trino — connectorx	14.66 s [14.51–14.71]	10.52 s	**617 MB**
Trino — ADBC driver	26.97 s [26.87–27.01]	18.64 s	1,963 MB
Trino — documented client	44.70 s [44.46–44.97]	24.05 s	4,455 MB

Figure 2 — S1 against every Trino client

SoftClient4ES · Trino clients



SoftClient4ES leads every route (1.23× vs the fastest).

connectorx builds contiguous Rust buffers and wins this one.

The counter-evidence is part of the result. Panels have independent scales.

> SoftClient4ES keeps the wall-clock lead against every Trino client (1.23× vs the fastest). **That is the tightest multiple in this report, so read it with its dispersion.** The two intervals do not touch — 11.73–12.01 s against 14.51–14.71 s across five runs each — and the known drift on our cell moves it the *other* way: measured settled, our side runs 10.04 s and the multiple becomes 1.46×. 1.23× is the conservative end of its own uncertainty, not the middle of it. On client cost the picture is honest: connectorx, building contiguous buffers in Rust, reaches a lower peak memory (617 MB) than our chunked Arrow batches, while still spending 3.7× our client CPU. **The wall-clock advantage comes from the server side — the paging strategy §7.1 isolates — and no client library on the row protocol recovers it** (connectorx parses the same stream in Rust and lands at 14.66 s). Note that 617 MB is a bare Arrow *table*: when the same workflow materialises the whole result as a DataFrame (scenario S5), connectorx needs 2.9 GB where SoftClient4ES needs 1.5 GB — so this memory advantage does not carry to the DataFrame an analyst actually keeps.

S1r Extract 10M rows into a DataFrame

The artifact a data scientist actually builds. Each side uses its most idiomatic route to a `pandas.DataFrame`, and separately to a `polars.DataFrame`.

```
# SoftClient4ES
df = cur.fetch_arrow_table().to_pandas()        # pandas
df = pl.from_arrow(cur.fetch_arrow_table())     # polars

# Trino
df = pandas.read_sql(sql, sqlalchemy_conn)      # pandas, via the trino SQLAlchemy dialect
df = pl.read_database(sql, sqlalchemy_conn)     # polars
```

To a pandas DataFrame

ROUTE	WALL	CLIENT CPU	PEAK MEMORY
SoftClient4ES (`to_pandas()`)	**9.97 s** [9.76–10.01]	**2.67 s**	1,481 MB
SoftClient4ES (`pd.ArrowDtype`)	9.95 s [9.94–10.17]	2.63 s	**919 MB**
Trino — connectorx	15.52 s [14.98–15.71]	11.54 s	2,824 MB
Trino — ADBC driver	27.29 s [27.08–27.42]	18.88 s	2,542 MB
Trino — SQLAlchemy (documented)	56.13 s [55.86–56.18]	35.30 s	8,079 MB

Against Trino's documented route that is **5.6× faster on 13.2× less CPU and 5.5× less memory**; against its fastest route, **1.6× faster on 1.9× less memory**. **Read those two multiples as a band, not a point.** Unlike S1, this cell was measured in the settled state (it ran later in the block sequence), so it carries the favourable end of the warm-in §2 describes; place our cell where S1's first block sat and the same comparisons read **4.7×** and **1.3×**. Trino's own position effect over the session is 0.4%, so the band is ours, not theirs. Both of our dtype backends are published rather than the flattering one: Arrow-backed dtypes cost nothing on wall clock this session (9.95 s against 9.97 s) and save 562 MB — the route a memory-bound analyst should take.

To a polars DataFrame

ROUTE	WALL	CLIENT CPU	PEAK MEMORY
SoftClient4ES (`pl.from_arrow`)	**10.94 s** [10.52–11.25]	11.31 s	2,484 MB
Trino — connectorx	15.34 s [14.89–15.45]	11.64 s	**2,186 MB**
Trino — ADBC driver	28.20 s [27.96–28.82]	28.17 s	3,551 MB
Trino — SQLAlchemy (documented)	56.04 s [55.85–56.34]	34.95 s	11,047 MB

Outcome. SoftClient4ES lands a DataFrame faster on every route and destination — **4.7–5.6×** faster than Trino's documented route on pandas and **4.3–5.1×** on polars, and **1.3–1.6×** (pandas) and **1.2–1.4×** (polars) faster than its fastest route; each band runs from our cell placed where S1's first block sat to the settled state it was actually measured in. On the polars destination both sides re-encode Arrow strings into polars' native format, which raises our client CPU; connectorx does the same re-encoding in Rust and comes in slightly lower on CPU and memory. The durable advantage, again, is wall-clock, held on every route.

S2 Extract 10M rows and aggregate in DuckDB

The same fetch, landed in DuckDB and aggregated. SoftClient4ES registers the Arrow table zero-copy; Trino builds a DataFrame from rows and registers that.

```
# SoftClient4ES
con.register("events", cur.fetch_arrow_table())          # zero-copy
con.execute("SELECT category, AVG(amount) FROM events GROUP BY category")

# Trino
df = pandas.DataFrame(cur.fetchall(), columns=cols)
con.register("events", df)
con.execute(same_agg)
```

METRIC	SOFTCLIENT4ES	TRINO	RATIO
Wall median	**10.32 s** [10.25–10.48]	52.95 s [52.86–53.39]	**5.1× faster**
Client CPU	**3.70 s**	32.45 s	8.8× less
Peak client memory	**944 MB**	7,510 MB	8.0× less

Outcome. The zero-copy Arrow hand-off to DuckDB is the largest memory advantage in the benchmark (8.0×): the columnar result is scanned in place instead of being copied through Python objects. Like S1r, this cell was measured in the settled state, so the wall-clock multiple is a band: **4.3× to 5.1×** depending on where our cell sits in the block sequence (§2, the warm-in). The memory ratio is unaffected — peak memory does not drift.

S3 `GROUP BY` returning 100 rows

`SELECT category, COUNT(*), AVG(amount) FROM bench_events_10m GROUP BY category`, returning 100 groups.

METRIC	SOFTCLIENT4ES	TRINO
Wall median	**0.043 s** [0.036–0.046]	5.50 s [5.42–5.61]
ES wire	**27.1 KB**	1.39 GB
Elasticsearch CPU	**0.1 s**	21.3 s
Rows	100	100

The same answer, not merely the same row count. A `terms` aggregation is approximate when its bucket size is too small, so "100 groups" is not by itself evidence that the pushed-down result matches a full scan. Every session gates on the values: all 100 `(category, COUNT, AVG(amount))` triples are compared across the three stacks — identical counts, averages equal to within 1e-9 — and Elasticsearch reports `doc_count_error_upper_bound = 0` and `sum_other_doc_count = 0`, its own statement that no bucket is missing and no document fell outside the returned buckets.

> **Outcome.** SoftClient4ES compiles the `GROUP BY` into an Elasticsearch `terms` aggregation: the cluster computes the 100 groups and returns only those 100 rows — 27.1 KB of aggregation response against the 1.39 GB Trino reads to compute the same answer, at 0.1 s of cluster CPU against 21.3 s and 0.043 s of wall against 5.50 s — the byte ratio is spectacular, but it is a ratio of bytes, not of time; the time columns are the claim. Trino's Elasticsearch connector performs predicate push-down only, so it scans all 10M rows into Trino and aggregates there.[1] **This is the clearest architectural difference in the benchmark** — for work Elasticsearch can do, SoftClient4ES does not move the data at all.
>
> **This is the one durable number in this report, and it reads differently from every other.** The extraction multiples are competitive results: they can erode with a Trino release, a tuning pass, different hardware — §7.1 shows our own headline moving 3.4× with one setting. This cell is a **property**: it follows from what the connector's documentation says it does not do, it is invariant to topology in the quantity that matters — the bytes that leave the cluster, 25.9 KB at one shard and 27.1 KB at six against Trino's 1.39 GB either way (§7; the wall-clock cells, 0.038 s and 0.043 s, are milliseconds and move with the session) — and it is validated answer-by-answer with `doc_count_error_upper_bound = 0`. It moves only if the connector itself changes. A reader who takes one number from this benchmark should take this one, not the 3.75×.

[1] The connector's own documentation for the version measured (`trino.io/docs/483/connector/elasticsearch.html`, consulted 2026-08-22) documents a *Predicate push down* section and no aggregation push-down of any kind; the 1.39 GB it reads off the cluster for a 100-row answer is this benchmark's measurement of that. It is the one load-bearing statement here that rests on a source rather than on a session artifact, so the source is named.

S1m Extract 1,000,000 rows — the only scale all three stacks reach

Elasticsearch's own query language cannot return more than 1,000,000 rows, so this is the largest result set on which SoftClient4ES, Trino and ES|QL can be compared like for like. Same eight columns, same index, `LIMIT 1000000` on every stack.

ROUTE	WALL	CLIENT CPU	PEAK CLIENT MEMORY	BYTES OFF THE CLUSTER
ES|QL, `format=arrow`	**0.32 s [0.28–0.34]**	0.05 s	165 MB	72 MB
ES|QL, `format=json`	0.98 s [0.95–1.07]	0.50 s	666 MB	86 MB
SoftClient4ES, Arrow Flight SQL	**3.99 s [3.97–4.05]**	**0.19 s**	176 MB	253 MB
Trino, documented client	4.42 s [4.40–4.46]	2.23 s	475 MB	308 MB

Outcome — ES|QL wins this scale; between the two SQL engines we are 1.11× faster. ES|QL remains an order of magnitude faster than anything else measured here, and that is the honest headline of this cell. Against Trino, SoftClient4ES lands the same million rows in 3.99 s against 4.42 s, on **11.7× less client CPU** (0.19 s against 2.23 s) and 2.7× less client memory. The reason ES|QL is fast is not the wire: it reads `doc_values`, columnar on disk, while both engines read `_source` and pay a JSON parse per document.

The wire explains the ordering. For the same million rows SoftClient4ES moves **253 bytes per row off the cluster against Trino's 308 bytes**, at the same cost per byte as its own ten-million-row run, which moves **254 bytes** per row. The million-row figure is the ten-million-row behaviour at one tenth the scale, not a different regime.

ES|QL Elasticsearch's own query language, and where it stops

A benchmark of two SQL engines over Elasticsearch that never measures what Elasticsearch itself can do invites an obvious question. So it is measured, over both of its wire formats: the row-shaped `format=json` every ES|QL client speaks, and `format=arrow`, which returns an Apache Arrow IPC stream.

SCENARIO	ES|QL (ARROW)	ES|QL (JSON)	SOFTCLIENT4ES	TRINO
S1m — 1,000,000 rows	**0.32 s [0.28–0.34]**	0.98 s [0.95–1.07]	3.99 s [3.97–4.05]	4.42 s [4.40–4.46]
S3 — `GROUP BY`, 100 rows	0.034 s [0.033–0.035]	**0.034 s [0.033–0.062]**	0.043 s [0.036–0.046]	5.50 s [5.42–5.61]
S4 — `LIMIT 100`	0.006 s [0.006–0.007]	**0.006 s [0.006–0.006]**	0.037 s [0.031–0.040]	0.065 s [0.056–0.067]
S1, S2, S5, S6 — 10,000,000 rows	cannot run	cannot run	✓	✓

All four columns are the same session as the sections above.

Where it wins. Below its ceiling ES|QL is the fastest way to get rows out of Elasticsearch that we measured: on a 100-row fetch its whole round trip (6 ms) costs less than SoftClient4ES's connection handshake alone (13.0 ms). Trino's handshake is cheaper still (1.6 ms); its 65 ms total is spent elsewhere. **Where it does not.** On the pushed-down aggregation the two are **level**: 0.034 s for ES|QL against our 0.043 s, both moving kilobytes rather than gigabytes off the cluster — 27.1 KB for us, 45.2 KB for ES|QL. At these magnitudes the difference sits inside this session's own noise. The work is identical; the difference is what the result travels over. **Why it is absent from four scenarios.** `esql.query.result_truncation_max_size` defaults to 10,000 rows and is declared with a hard maximum of 1,000,000; Elasticsearch refuses a higher value outright. Ask for ten million rows with the setting at its maximum and the response is 1,000,000 rows, HTTP 200, and no `Warning` header — a truncated answer that looks like a complete one, recorded as `esql-truncation-probe.json` in the session directory. Not a licence gate: the cluster measured here runs a `basic` licence. For result sets under a million rows, ES|QL is an excellent answer and this benchmark is not the argument for anything else. **The second boundary is the join.** `LOOKUP JOIN` does not join two fact indices: it resolves only against an index in `lookup` mode, and such an index is **restricted to a single shard** — a dimension table. Asked for this benchmark's cross-index join it answers `400 — invalid [bench_1m] resolution in lookup mode to an index in [standard] mode`, so section 6's joins have no ES|QL column either: not that it is slower, but that the shape is outside what the feature accepts.

S4 Fetch 100 rows (`LIMIT 100`)

The control: when the result is small, the client representation stops mattering.

METRIC	SOFTCLIENT4ES	TRINO
Wall median	0.037 s [0.031–0.040]	0.065 s [0.056–0.067]

Outcome. Near parity, as expected. The extraction advantage only appears at scale, or when work can be pushed into Elasticsearch.

What the 32 ms actually is, and a control that removes it. At this size the whole scenario is connection setup: SoftClient4ES's ADBC Flight SQL handshake takes 13.0 ms of the 37, Trino's 1.6 ms of the 65. Both clients dial an IP literal here, deliberately — because when SoftClient4ES is pointed at a hostname instead, its connect cost rises to 30.5 ms and the S4 median moves to **0.054 s**, closing most of the distance to Trino's 0.065 s. The margin on this control is therefore worth about as much as a name lookup. The cause is client-side and measured: a four-layer probe over 30 repeats separates a bare TCP connect (0.07 ms) from the Flight C++ layer (1.64 ms), the Go ADBC layer dialling an IP (3.01 ms) and the same layer dialling a name (16.07 ms). grpc-go performs its own resolution rather than using the OS resolver, which is where the ~15 ms goes. No other scenario is affected — 15 ms is invisible against 10 seconds — but on a 100-row fetch it is most of the result.

Correctness gate — run before any of the timings above. Row counts alone do not establish that a pushed-down aggregation returns the same answer as a full scan, so each session first runs a data-equivalence gate, and a session whose gate does not pass is not measured at all. **All three stacks take part in both halves.** On the push-down itself — every one of S3's 100 `(category, cnt, AVG(amount))` triples, key by key and value by value: they agreed exactly on the counts and to within 1e-9 on the averages. And on `COUNT`, `SUM(id)`, `SUM(qty)` and `SUM(amount)` over the whole index — the check that the extraction paths see the same 10,000,000 documents, not merely the same number of them: all three agreed on all four. Elasticsearch's own error bounds for the `terms` aggregation were `doc_count_error_upper_bound = 0` and `sum_other_doc_count = 0` — the aggregation is exact at this shard count and bucket size, not merely close.

5 Constrained memory and concurrency

S5 Does the extraction fit in a small container?

The client runs inside a container with a hard memory cap (`docker run --memory`, swap pinned equal so the kernel kills rather than pages). The question is binary: does landing 10M rows as a DataFrame complete, or is the process killed?

Whole result set as one DataFrame

CONTAINER CAP	SOFTCLIENT4ES	TRINO (DOCUMENTED)	TRINO (CONNECTORX, FASTEST)
8 GB	completes 10.9 s · 1,537 MB	OOM-killed	completes 13.1 s · 2,911 MB
6 GB	completes 11.3 s · 1,535 MB	OOM-killed	completes 13.4 s · 2,909 MB
4 GB	completes 10.7 s · 1,533 MB	OOM-killed	completes 12.6 s · 2,912 MB
3 GB	completes 10.7 s · 1,541 MB	OOM-killed	completes 13.4 s · 2,909 MB
2 GB	completes **11.4 s · 1,531 MB**	OOM-killed	OOM-killed

The whole table comes from one sweep per client. "OOM-killed" is the kernel killing the client — exit 137 — in every cell above, verified per cell.

Streaming, where neither side holds the whole result — SoftClient4ES via `fetch_record_batch()`, Trino via `pandas.read_sql(chunksize=…)`. This is the workflow a competent Trino user reaches for first, so it is a table rather than a footnote:

CONTAINER CAP	SOFTCLIENT4ES	TRINO (DOCUMENTED)
8 GB	completes 10.9 s · 156 MB	completes 58.5 s · 312 MB
6 GB	completes 10.8 s · 153 MB	completes 58.5 s · 313 MB
4 GB	completes 10.8 s · 154 MB	completes 58.8 s · 314 MB
3 GB	completes 10.7 s · 155 MB	completes 59.2 s · 315 MB
2 GB	completes **10.8 s · 153 MB**	completes **58.6 s · 312 MB**

Both complete at every cap. SoftClient4ES is **5.4–5.5× faster on 2.0× less memory** — a smaller and more representative difference than the binary one above, and the one that applies whenever the workflow does not need the whole result set in memory at once.

Outcome. Landing the whole 10M-row DataFrame, SoftClient4ES fits in a **2 GB** container. Trino's documented client is killed at every cap measured, 8 GB included; its fastest client (connectorx) fits at 3 GB but is killed at 2 GB — the cap where SoftClient4ES still completes. Both requirements are independent of the cap (peak memory moves by under 1% across a 4× range), so they reflect the data, not memory thrashing. The binary framing is the harder claim, and it answers "can I materialise this result in a small container", not "can this pipeline run at all" — read it next to the streaming table above, where both engines run everywhere and we are **5.4–5.5× faster on 2.0× less memory**. This session's streaming arm is flat — 10.7–10.9 s across a 4× cap range at 153–156 MB — where a previous campaign's wandered by 40%; the noise did not reproduce.

S6 Concurrent extractions in a fixed memory budget

An 8 GB total client budget, split evenly across N clients launched simultaneously, each extracting 10M rows. How many complete with the correct row count? Measured against **both** of Trino's clients — the documented one it ships and connectorx, the fastest it has, which S1 already grants it.

ENGINE	N	CAP EACH	COMPLETED	WALL
SoftClient4ES	1	8,192 MB	1 / 1	11.2 s
SoftClient4ES	2	4,096 MB	2 / 2	17.7 s
SoftClient4ES	3	2,730 MB	3 / 3	25.0 s
SoftClient4ES	4	2,048 MB	4 / 4	34.4 s
SoftClient4ES	**5**	**1,638 MB**	**5 / 5**	**42.8 s**
Trino — connectorx	1	8,192 MB	1 / 1	13.6 s
Trino — connectorx	**2**	**4,096 MB**	**2 / 2**	**21.0 s**
Trino — connectorx	3	2,730 MB	0 / 3	killed
Trino — documented client	1	8,192 MB	0 / 1	killed

One run per level, like the S5 sweeps: the measured outcome is how many clients complete with the correct row count, not a wall-clock median, so these rows carry no dispersion.

Outcome. In an 8 GB budget, SoftClient4ES completes **five** concurrent extractions — 50 million rows — against **two** for Trino's fastest client and none for its documented one. The ratio is 2.5×, not the five-versus-nothing that client alone would suggest, and it follows directly from what each client holds per extraction. **Concurrency is not free, and the reason is ours.** Five extractions take 42.8 s against 11.2 s for one — 3.8× the time for 5× the work. Each of our clients opens six concurrent slices, so five of them put **thirty readers on a six-CPU Elasticsearch cluster**: sliced paging spends cluster budget, and at N=5 the cluster, not the client, is what sets the pace. The claim is capacity, not scaling. This measures *client-side* capacity; it does not measure server-side concurrency, which is a separate question this benchmark does not address.

6 Cross-index JOIN

A join between a 1M-row index (`bench_1m`) and the 10M-row index, landed as a pandas DataFrame on both sides. Both engines execute the join server-side (SoftClient4ES in an embedded DuckDB, Trino in its own engine).

```
-- J0  plain join                    (1,000,000 rows out)
SELECT a.id, b.amount FROM bench_1m a JOIN bench_events_10m b ON a.id = b.id
-- J1  + predicate on the large leg  (125,044 rows out)
SELECT a.id, b.amount FROM bench_1m a JOIN bench_events_10m b ON a.id = b.id WHERE b.status =
'paid'
-- J2  + GROUP BY                     (100 rows out)
SELECT b.category, COUNT(*), AVG(b.amount) FROM bench_1m a JOIN bench_events_10m b ON a.id =
b.id GROUP BY b.category
```

SCENARIO	SOFTCLIENT4ES WALL	TRINO WALL	CLIENT CPU (SC4ES / TRINO)
J0 — plain join	7.99 s	**7.43 s** (7.0% faster)	0.05 s / 1.16 s
J1 — + `WHERE`	3.93 s	**3.44 s** (12.4% faster)	0.02 s / 0.19 s
J2 — + `GROUP BY`	8.56 s	**6.66 s** (22.3% faster)	0.01 s / 0.05 s

Medians of 5 runs, as everywhere else. These are small differences, so the spread belongs next to them — and each engine's 5 runs are compared against the other's 5, 25 pairings per scenario, so the ordering does not rest on the medians alone:

SCENARIO	SOFTCLIENT4ES MIN–MAX	TRINO MIN–MAX	RUNS WON, OF THE 25 PAIRINGS
J0	7.85–8.16 s (3.9%)	7.41–7.63 s (2.9%)	**Trino 25 / 25**
J1	3.91–3.95 s (1.1%)	3.39–3.50 s (3.1%)	**Trino 25 / 25**
J2	8.53–8.62 s (1.0%)	6.52–7.03 s (7.6%)	**Trino 25 / 25**

Outcome — Trino wins all three. By **7.0%** on the plain join, **12.4%** with a predicate and **22.3%** with an aggregation, winning **every one of the 25 pairings in every scenario**; the ranges do not touch anywhere. Its advantage *grows* with the work added to the join: adding the `GROUP BY` of J2 costs Trino **less than nothing** (J2 is 0.77 s faster than its own J0) against **+0.57 s** for us. The one axis that runs our way is client cost — **5–23× less client-side work** — but on a join returning 100 rows that is a small absolute saving and does not offset the wall clock. J0 and J2 come from one block in which both engines were measured together, warm — all three scenarios in one counter-balanced block, both join legs pre-warmed. An earlier block in which Trino's J2 degraded 37% across its own runs is quarantined, and discarding it made Trino's win larger.

7 Index topology sensitivity

Everything above was measured on a 6-shard index. Trino's Elasticsearch connector creates one split per shard, so the natural question is how much of the gap is shard parallelism — and whether either engine actually exploits it.

Both do. They do not exploit it equally.

> **Setup.** Same host, same engines, same corpus regenerated from the same seed into a **1-shard** index. **Only the shard count changes**: the engines, their resources and the 1.5×-CPU handicap in Trino's favour are the ones stated in section 2, unchanged. Medians of 5 runs after 2 warm-ups. The parallelism under test was verified rather than assumed: a live probe of `system.runtime.tasks` during the scan recorded **6 splits on the 6-shard index, 3 on each worker and none on the coordinator, against 1 split on the other, none on the coordinator**.

METRIC	SOFTCLIENT4ES 1 SHARD	SOFTCLIENT4ES 6 SHARDS	TRINO 1 SHARD	TRINO 6 SHARDS
S1 wall	41.69 s [40.93–65.60]	**11.91 s [11.73–12.01]**	55.11 s [54.18–56.17]	44.70 s [44.46–44.97]
S1 client CPU	3.56 s	**2.83 s**	24.29 s	24.05 s
S1 peak client memory	922 MB	**921 MB**	4,473 MB	4,455 MB
S1 Elasticsearch CPU	25.4 s	42.2 s	28.7 s	30.4 s
S1 ES wire	2.53 GB	2.54 GB	2.96 GB	2.92 GB
S3 wall (`GROUP BY`, 100 rows)	0.038 s [0.033–0.045]	**0.043 s [0.036–0.046]**	28.54 s [28.42–29.73]	5.50 s [5.42–5.61]
S3 ES wire	25.9 KB	**27.1 KB**	1.43 GB	1.39 GB

> **Outcome — extraction.** SoftClient4ES goes **3.50× faster** with six shards (41.69 s → 11.91 s). Trino goes **1.23× faster** (55.11 s → 44.70 s), with six real splits across both workers. The S1 ratio therefore **widens from 1.32× to 3.75×**. On one shard the two engines are close: nearly all of the headline gap is shard parallelism, which §7.1 attributes by switching concurrent paging off on the same build rather than by comparing builds.

Why Trino gains so little from six splits. Not because it fails to parallelise — the probe shows six splits on two workers. Its wall clock is bound by its own row processing rather than by how fast Elasticsearch can serve pages, and the server-side counters say so: Trino's Elasticsearch CPU barely moves between topologies (28.7 s → 30.4 s) while its engine CPU stays near 56 s in both. Ours moves the other way — Elasticsearch CPU **rises** 25.4 s → 42.2 s, because six concurrent readers drive the cluster harder to finish in a quarter of the time.

What this establishes, and what it does not. Elasticsearch here is a **real 3-node cluster** (3 × 2 CPU), not one container wearing three hats, so the six splits and six slices really are served by three independent nodes — the result is about shard parallelism, not about a single node being the bottleneck. What remains untested is a **larger** cluster — more nodes, more shards, and shards per node above two; a reader operating one should treat the scaling *slope* as measured on three nodes rather than as a general law.

Client memory is a property of the protocol, not the topology — client CPU is not. Peak client memory moves by **0.2% on our side and 0.3% on Trino's**: the client is one process consuming one wire format however large the cluster is. Client *CPU* does move on our side — **3.56 s on one shard against 2.83 s on six**, while Trino's is flat (24.29 s → 24.05 s). Six slices deliver the same bytes in more, smaller, concurrently-assembled batches, so the per-batch work overlaps the wait instead of queueing behind one reader; the wire format is unchanged, the scheduling is not. Section 4's constrained-container result was **not** re-measured on the 1-shard index, so the 2 GB outcome there is an inference from the flat memory column, not a measurement.

Pushdown is architectural. The `GROUP BY` still moves **27.1 KB off the cluster against 1.39 GB** — the 100-row answer itself. Six shards do not change it — the Trino cluster is identical in both columns — because the connector pushes no aggregation down: it reads all 10 million rows whatever the topology, and merely reads them faster.

Where Trino gains most

Its aggregation wall-clock improves **5.2×** (28.54 s → 5.50 s), by far the largest single improvement measured in this benchmark for either engine. Scan parallelism is exactly what a multi-shard index gives Trino, and on aggregate-heavy work it converts directly into wall-clock.

Both ES-wire columns in this section are measured at the Elasticsearch container itself — bytes that left the cluster — which matters here because with a three-node Trino the coordinator's own counter sees only a fraction of the scan.

7.1 What the parallelism is worth — sliced paging measured against itself

Section 7 shows the gap widening with shards. This subsection attributes it, by holding **everything** constant except one setting on our own side.

SoftClient4ES pages a no-`ORDER BY`, no-`LIMIT` extraction with `min(primary shards, max-slices)` concurrent point-in-time slices. The shipped default `max-slices = 8` resolves to 6 on this index; setting it to **1** restores a single sequential reader. Same image, same cluster, same corpus, minutes apart. Which branch each arm took was **observed, not assumed** — the sidecar reports its slice count per extraction, and each arm asserts on it before measuring:

```
sequential arm:  PIT search_after completed (1 slice(s))
sliced arm:      PIT search_after completed (6 slice(s))
```

CELL	SEQUENTIAL (1 SLICE)	SLICED (6 SLICES)	GAIN
S1 — extract 10M rows	35.34 s [35.09–39.42]	**10.48 s [10.14–10.58]**	**3.37×**
S2 — 10M rows into DuckDB	36.97 s [36.06–38.76]	**14.49 s [13.31–15.43]**	2.55×
S3 — `GROUP BY` (control)	0.03 s	0.03 s	—
S4 — `LIMIT 100` (control)	0.02 s	0.02 s	—
S1 engine CPU	21.4 s	25.7 s	+20%
S1 **Elasticsearch CPU**	53.1 s	**37.4 s**	**−30%**

Read this table across, not against §4. Each arm is measured minutes after its pair, on a cluster that pair has just worked hard, so the absolute values here do not line up with the matrix — and they miss it in *both* directions: the sliced S1 arm lands **12% below** §4's 11.91 s, while the sliced S2 arm lands **40% above** §4's 10.32 s, carrying 44% more Elasticsearch CPU than the matrix cell did (51.0 s against 35.5 s) — the state its own sequential arm left behind. That is why the claim of this section is a *ratio within a row* — same image, same corpus, one setting different — and not a wall-clock figure to be quoted beside §4's.

The controls are the point of the table. S3 and S4 do not page — one is an aggregation, the other returns 100 rows — and both are identical to the hundredth of a second *across the two arms* (0.03 s and 0.02 s), which is the comparison this section makes. They are not identical to §4's 0.043 s for the same S3 fetch, for the same session reason as above. Whatever moved S1 and S2 did not move them, so the arms differ by the paging strategy and nothing else.

It is less work, not just more overlap. Elasticsearch CPU **falls 30%** while our engine CPU rises 20%. Sequential paging issues ~10,000 round trips that each fan out to all six shards and merge — roughly 60,000 shard operations; sliced paging sends each page to one shard. So the 3.37× is not parallelism hiding latency behind more total work: the cluster does measurably less of it.

Scope. Slicing applies to extractions without `ORDER BY` or `LIMIT`, which is the shape of every large extraction in this report. An ordered or bounded query still pages sequentially, and a 1-shard index cannot slice at all — `min(1, 8) = 1` — which is the other reason section 7's 1-shard column is what it is.

8 Calibration, and where Trino is stronger

Calibration — reference points, not alternatives.

Two things in this report extract faster than either SQL engine. Neither is a candidate for the workload this benchmark is about: one stops at a million rows and cannot join two fact indices, the other is a hand-maintained Query DSL document. They are measured and published because a reader who knows Elasticsearch will ask about them, and an unmeasured question reads as an avoided one. They calibrate the scale; they do not compete on it.

- **Up to one million rows, Elasticsearch itself extracts faster than either engine** — ES|QL returns 1M rows in **0.32 s** over Arrow against SoftClient4ES's 3.99 s and Trino's 4.42 s, and a 100-row fetch in 6 ms against 37 ms (section 4). Its 1,000,000-row ceiling and its refusal to join two fact indices are why it is absent from the rest of this report. On the segment it does cover — result sets under a million rows, no joins, no SQL — it is a credible answer, and this report is not the argument against it.
- **ES|QL ties us on the pushed-down aggregation.** The two are level — **0.034 s against our 0.043 s**, and their five-run intervals overlap (0.033–0.062 s against 0.036–0.046 s). The honest statement is a tie. The gap on extraction has a named cause and a price: ES|QL reads columnar `doc_values` while both SQL engines read `_source` and pay a JSON parse per document. The whole S1m gap between its Arrow route and ours is **3.67 s per million rows** — an upper bound on what that leg can be worth, since the two routes also differ in where they run. It is the largest identified item left on our extraction path: a roadmap line, not a defeat.
- **A hand-written sliced scroll is the extraction floor** — six processes, one slice per shard, building the same Arrow table: 13.76 s against our 11.91 s (section 4). It answers one query shape, in JSON rather than SQL, at 8.5× our client CPU and 4.0× our client memory.

Where Trino is stronger.

- **Aggregation over a sharded index** — given shards to parallelise over, Trino's `GROUP BY` wall-clock improves **5.2×** (28.54 s → 5.50 s going from 1 shard to 6), the largest single gain either engine showed in this benchmark. It still moves 1.39 GB off the cluster to get there.
- **Cross-index joins — Trino wins all three**, and by clear margins: **J0 by 7.0%**, **J1 by 12.4%** and **J2 by 22.3%**, winning all 25 pairings in each. It is also more efficient at aggregating over a join: adding the `GROUP BY` costs Trino less than nothing (J2 is 0.77 s faster than its own J0) against +0.57 s for us.
- **A 100-row fetch is a tie once a name lookup is involved** — our 28 ms margin on S4 is worth about one hostname resolution in our gRPC client: dialling a name instead of an IP costs us 17 ms and moves the median to 0.054 s against Trino's 0.065 s (section 4). grpc-go resolves names itself in ~15 ms rather than using the OS resolver.
- **Its page size is not tuned in these results, and tuning it helps a little** — raising `elasticsearch.scroll-size` from its default 1000 to 5000 improves Trino's S1 wall clock from 44.70 s to **44.41 s — 0.7%**. On this cluster the knob has almost nothing left to give, which is consistent with Trino being bound by its own row processing rather than by paging. The headline tables pair product default against product default.
- **A lower-memory Arrow client exists, and it is a real win** — via connectorx, Trino lands an Arrow *table* in **617 MB against our 921 MB**, 33% less client memory than we use, and it is the closest client to us on wall clock (14.66 s against 11.91 s). Anyone quoting a multiple from this report should

quote it against this route. (It does not extend to every destination — S5, where connectorx needs 3 GB and we complete in 2 GB.)

- **Distributed execution, spill-to-disk and fault tolerance** — Trino scales a single query across a cluster far larger than the 3 nodes it runs on here, and spills to disk; this benchmark does not exercise any of it.
- **Connector breadth** — Trino federates 40+ data sources; SoftClient4ES is Elasticsearch-focused.
- **Licensing** — Trino is Apache-2.0 throughout; SoftClient4ES gates result-set size by licence tier (see Reproduction).

9 Reproduction

Prerequisites: Docker (VM ≥ 13 CPUs / ≥ 32 GB RAM — the measured configuration), a native CPython 3.12, and ≥ 32 GB host RAM — the client runs on the host, and Trino's documented client materialises 10M rows into Python objects.

```
python3.12 -m venv .venv
.venv/bin/pip install -r requirements.txt

# Bring up Elasticsearch and load the data (deterministic; ~3 minutes)
docker compose up -d elasticsearch
.venv/bin/python generator/generate_and_load.py

# Bring up both engines
docker compose up -d flight-sql trino

# The published matrix is three scripted phases, not four commands: 180 measured
# runs over ~16 orchestrate invocations, every arm tagged so it cannot blend into
# another's median. Medians of 5 runs after 2 warm-ups throughout.

# Phase 1 — floors S0/S0p; S1/S1m/S1r/S2/S3/S4 on both engines; Trino's connectorx
# and ADBC routes; every S1r destination on both sides; ES|QL on both wire formats
# plus its truncation and join probes; the tuned-Trino (scroll-size=5000) and
# hostname-dial arms; the four-layer connect probe; the drift control, last.
# Refuses to measure unless the equivalence gate PASSes, then WARMS THE CORPUS to a
# verified steady state (warm-in.json) before the first timed block.
/bin/bash runners/run_full_session.sh

# Phase 2 — S5 memory cap and S6 concurrency, each on two client routes, then
# J0–J2 behind wait_engines_idle.py (a join block once timed on top of Trino's
# unfinished work and was discarded; the gate is why that cannot recur).
/bin/bash runners/run_full_session_phase2.sh

# Phase 3 — section 7: the corpus regenerated from the same seed into 1 shard,
# its own equivalence gate, and a live system.runtime.tasks split probe.
/bin/bash runners/run_full_session_phase3.sh

# Re-derive every published figure from the artifacts; non-zero exit on any
# disagreement between the documents, the report and the runs.
.venv/bin/python runners/verify_claims.py
```

Results are written under `results/<session>/`, one JSON per run, with the environment and the sidecar image digest captured automatically. `session.log` is the authoritative record of what ran, in order, with timestamps.

Licence — what reproduces at which tier. SoftClient4ES enforces a *result-set* quota by licence tier (Community 10,000 rows / Pro 1,000,000 / Enterprise unlimited). **The licence lifts a row quota only — it does not change the batch size, the serialization, or the data path**, and the runners assert exact row counts, so a quota-capped run aborts rather than being reported: a truncated result can never be published by accident. The quota applies to the rows *returned*, not the rows read, which makes the free reach larger than the tier names suggest.

- **With no licence at all (Community). S3** — the push-down, and the one number this report asks a reader to take away — returns 100 rows from an aggregation the cluster computes, and aggregation-shaped queries carry no row cap at all. **S4** runs, and so does **J2** (100 rows out, one join against a Community allowance of two). Every other scenario reproduces at reduced `--limit` scale. **The structural result is verifiable without asking anyone for anything.**
- **With the 30-day Pro trial** — self-service from the licence portal, no credit card, one per account: **S1m** (1,000,000 rows) and the cross-index joins — **J1** (125,044 rows out) and **J0**, whose 1,000,000-row output sits exactly on the Pro ceiling, which the cap admits because it truncates only *above* the quota. That covers every cell where Trino wins or ties.
- **Enterprise** — the ten-million-row extraction cells: S1, S1r, S2, S5, S6. A **time-limited evaluation licence** for those, with no contractual obligation attached, is requested from your account on the pricing page of the licence portal.

portal.softclient4es.com/pricing

A Appendix — Methodology

What this benchmark measures — and what it does not

The data path under test:

```
Elasticsearch → row-serialized documents out of the cluster (the _search / _sql path
                both engines read; ES|QL's columnar Arrow output is a third path,
                measured separately and capped at 1,000,000 rows)
             → SoftClient4ES: converted to Arrow ONCE at the sidecar, streamed as
               Arrow batches, consumed by the client with no further deserialization
             → Trino: parsed into Trino's columnar pages, re-serialized to Trino's
               row-oriented client protocol, re-parsed value-by-value in the client
```

Every client pays an Elasticsearch serialization leg. Nothing leaves the cluster unserialized, and this benchmark never claims to avoid that step. For the SQL and JDBC path it is about, that serialization is **row-shaped**: `_search` returns JSON, and the SQL endpoint offers CSV, TSV, YAML and the binary CBOR and Smile — every one of them a row-serialized document encoding.

One Elasticsearch format is columnar, and it is measured here rather than glossed over: ES|QL's `format=arrow` returns an Apache Arrow IPC stream (verified on Elasticsearch 8.18.3), and below its ceiling it *extracts* faster than either engine in this benchmark — though not on the pushed-down aggregation, where the two are **level** — **0.034 s** for ES|QL against our **0.043 s**. Both of its wire formats land on the same median there, so best-route discipline costs nothing to apply: the number is 0.034 s either way. At that magnitude the two five-run intervals overlap — ES|QL 0.033–0.062 s, ours 0.036–0.046 s — so section 8 records it as a tie. That ceiling is why it appears in three scenarios and not in the rest: `esql.query.result_truncation_max_size` defaults to 10,000 rows and is declared with a hard maximum of 1,000,000, so ES|QL cannot return a ten-million-row result at all — and above the configured ceiling it truncates with HTTP 200 and no `Warning` header.

What this benchmark measures is what happens *after* the serialization leg, on the path a SQL client takes: SoftClient4ES serializes to Arrow once and the client consumes that columnar buffer directly, while Trino re-serializes to a row protocol the client must parse again.

So this benchmark measures **large result-set extraction and client-side consumption** — wall-clock, client CPU, peak client memory — which is where the client representation dominates. It does **not** measure Elasticsearch-side query execution (I/O, scoring), where the wire format is irrelevant.

Fairness rules

- **Same host, same session, same index.** The whole 6-shard matrix is one session (one gate, one host state, 180 measured runs); S5, S6, the joins and section 7's 1-shard arm each rebuild their own container topology or index, so each is its own session, run the same morning on the same host and image. Every stack measured in a scenario — SoftClient4ES, Trino, and ES|QL where it can run — reads `bench_events_10m` back to back.
- **Container limits, and the handicap they encode:** Elasticsearch and the sidecar each get 4 CPU / 4 GB; **Trino gets more, deliberately** — a 3-node cluster totalling 6 CPU / 8 GB, in every scenario. Elasticsearch heap is pinned at 2 GB; Trino's official image auto-sizes its heap from container memory.
- **Page-size symmetry:** the sidecar's `ARROW_BATCH_SIZE=1000` equals Trino's `elasticsearch.scroll-size=1000`. Both are product defaults, pinned explicitly so the setting can be checked.
- **Authentication disabled on both paths**, so authentication cost is zero and identical on both sides.

- **Both clients are used the way their projects document them.** SoftClient4ES is driven with the standard `adbc_driver_flightsql` Arrow Flight SQL driver; Trino with its own `trino` Python package, and — for the fairness comparison in S1/S1r — with its fastest available clients (connectorx and the ADBC Trino driver). No SoftClient4ES-specific client code is used.
- **Trino uses its default type mapping.** `legacy_primitive_types` would return unparsed strings, deferring the parse rather than avoiding it, and is not used.
- **Warm cache, fresh process.** Each scenario's warm-ups run immediately before its measured runs, so both systems measure against a warm Elasticsearch page cache. Every measured run is a fresh OS process, so peak memory cannot leak between runs. Because SoftClient4ES runs first in each scenario, Trino meets a warmer cluster. Section 2 measures what that is worth on the headline cell and it is **not** slight: re-running S1 for both engines at the end of the session moved us 16% (11.91 s → 10.04 s) and Trino 0.4%. The effect runs **against us**, and this report publishes the **first-measured block** as the headline, printing the end-of-session pair as the favourable bound it does not use.
- **Correctness before timing.** Each run asserts the exact expected row count; a run that returns the wrong count is discarded, never reported. Python is never run with `-O`, so the asserts always execute.
- The exact configuration lives alongside the harness, and the data generator is deterministic (fixed seed).

Metrics

- **Wall-clock:** `time.perf_counter()` around connect → materialized result. Connection setup is included because it is time the user waits. **Every stack dials an IP literal**, so no arm resolves a name. Trino's Python clients resolve through the OS resolver; the Go ADBC driver uses grpc-go's own, which ignores `/etc/hosts`. A four-layer connect probe separates the cost: bare TCP 0.07 ms, the Flight C++ layer 1.64 ms, the Go ADBC layer dialling an IP 3.01 ms, and **the same layer dialling a name 16.07 ms** — a ≈13 ms resolver tax that belongs to the driver, not the protocol, with a 5 s DNS timeout behind it (softclient4es-arrow#151). The scale decides whether that matters: invisible against a ten-million-row extraction (13 ms against 11.91 s), but the whole result on the 100-row control. Both dials are measured and both published: by IP, S4 is 0.037 s against Trino's 0.065 s; by name it is 0.054 s. The name lookup does not invert that control, but it closes most of it. Published as a deployment note — with the Go driver, dial an address or reuse the connection.
- **Client CPU:** `time.process_time()` for the client process — the CPU spent turning the wire format into usable values.
- **Peak client memory:** the process's peak physical footprint. On macOS this is the kernel's `ri_lifetime_max_phys_footprint`, which includes compressed pages and is therefore immune to the macOS memory compressor (plain RSS under-reports under memory pressure). On Linux (the §5/§6 containers) the harness falls back to `ru_maxrss` — a different instrument, which is why §4 and §5 memory figures are never quoted on one scale.
- **ES wire bytes:** read from the container network counters, so a wire-volume difference is measured, not asserted. The counter is read **at the Elasticsearch container** in every table of this report, section 7 included, so the figure cannot depend on how many containers the engine is made of.
- **Server-side cost: measured on every cell of the main session, published where it carries a claim.** Engine and Elasticsearch CPU are read per run from each container's cgroup (`cpu.stat` / `usage_usec`, an exact counter; the throttling counters come from the same file) for **the whole 6-shard matrix** — floors, S1, S1m, S1r, S2, S3, S4 — and for §7's topology arm and §7.1's paging A/B. The tables print it where it carries a claim (the floors, S1, S3, §7, §7.1); on the remaining cells it is in the session records rather than in the prose, and §7.1's cross-reference to S2's 35.5 s comes from there. **Three groups are the exception**: S5, S6 and the joins rebuild container topologies mid-run, so they stay client-side only — reading *those* rows as total system cost is unsupported. The push-down is *not* credited with bytes alone: S3 publishes its cluster CPU (0.1 s against 21.3 s) beside the bytes precisely so it does not have to be.

Interpretation guardrails

- **S1 / S1r / S2** differences are attributable to client-side representation and the paging strategy (§7.1) — not to wire volume, which §4 shows running *in Trino's favour* on the client leg.
- **S3** differences are attributable to **aggregation pushdown**, not the wire format: SoftClient4ES compiles the `GROUP BY` into an Elasticsearch aggregation, while Trino's connector performs predicate pushdown only and scans all rows. S3 is never a wire-format result.
- **S4** is the control: with a small result the wire format stops mattering, and near-parity there is a sign the benchmark is honest, not a weakness.
- **Where Trino is stronger** is published alongside the results (section 8).

Known limitations

- **Single host throughout — but not a single node.** Trino runs as a real 3-node cluster (coordinator plus two workers, `node-scheduler.include-coordinator=false`), so its distributed execution *is* exercised; what is not, is scale-out across machines, over a real network, with spill-to-disk or fault tolerance in play. Elasticsearch is also a real 3-node cluster (3 × 2 CPU), so the splits and slices are served by independent nodes; what is untested is a *larger* cluster. Those are real strengths of Trino that an extraction benchmark on one host cannot show.
- **The Docker VM's 13 CPUs are oversubscribed by the declared limits (16), with asymmetric headroom per arm** — our arm claims 10 of 13 (sidecar 4 + Elasticsearch 6), Trino's 12 of 13 (Trino 6 + Elasticsearch 6). Limits are not reservations, so nothing fails — but the per-run cgroup throttling counters (`nr_throttled` / `throttled_usec`, in the same `cpu.stat` file the CPU figures come from) that would settle whether any container hit its limit during a run **are captured per run in this session, for every stack**, and §2 publishes the verdict: zero throttled periods for either engine and for Elasticsearch on every 10M-row cell, with Trino's single 0.8 s blip during the 4.4 s S1m burst. The residual caveat is the VM-level oversubscription itself, which cgroup counters cannot see; the load average and the flat swap band bound it.
- **Sliced paging buys wall clock with cluster CPU, and the peak is what a shared cluster feels.** Our headline cell costs Elasticsearch **42.2 s of CPU over an 11.91 s run** — an average of **3.5 of the cluster's 6 CPUs** for its duration, against 0.68 for Trino's documented client and 2.2 for connectorx. Six concurrent readers finish sooner by asking the cluster for more at once. On a cluster that also serves search traffic that is a real operational trade-off, and nothing here measures what it does to a concurrent search workload; S6 measures only what it does to our own concurrency (five clients put thirty readers on six Elasticsearch CPUs, and the cluster becomes the limit). The counterweight is measured and published in §7.1: over a whole extraction, sliced paging costs the cluster **30% less** total CPU than sequential paging, because each page goes to one shard instead of fanning out to all six. Higher peak, lower integral — and the peak is the half an operator has to size for. `elastic.scroll.max-slices = 1` turns it off.
- **The index is force-merged and carries no replicas.** Both engines read the same idealised corpus — one segment per shard, no replica competing for page cache, nothing writing to the index during a run. That removes a real source of variance and flatters both sides equally, but nothing here describes either engine on a live index under ingest, or on a cluster where replicas serve part of the read.
- **Slicing is measured at one point on its own curve.** The shipped default `max-slices = 8` is never reached here: the index has 6 primary shards, so every sliced run uses 6. What happens past the default — 12 or 24 shards against a cap of 8 — is unmeasured, so the 3.37× of §7.1 should not be extrapolated beyond the point where the default saturates.
- **The push-down generalisation rests on one aggregation shape** — a single-level `terms` on a keyword field with `COUNT` and `AVG`. `date_histogram`, multi-level `GROUP BY`, `HAVING`, and above all `COUNT(DISTINCT)` are unmeasured — the latter matters because it maps to Elasticsearch's `cardinality`

aggregation, which is *approximate by construction*, where Trino's `count(distinct)` is exact: a correctness divergence the equivalence gate does not yet test (it gates `COUNT`, `SUM`, `AVG`). Extending the gate and the matrix to it is queued.

- **One flat mapping, no nested fields or arrays.** This avoids Trino's documented gaps in those types, so the connector is not handicapped; it also means the benchmark says nothing about SQL coverage, only about extraction efficiency.
- **One column shape: eight columns over five types** (`long`, `date`, `double`, `integer`, `keyword`). The client-side cost of a wire format is a function of column count and type mix, and only one point on that curve is measured. Nothing here establishes how the gap moves for two columns or eighty.
- **Server-side cost is a recent instrument, and three groups still lack it.** Until 2026-08-19 this harness measured the client alone, which left "Trino holds half again our CPU" resting on an allocation rather than a consumption, and left the floor comparison lopsided — a sliced scroll pays everything in its client processes while SoftClient4ES also pays a sidecar JVM nobody counted. Both are now measured, from cgroup `usage_usec` read inside each container (the engine's containers summed — Trino is three — and Elasticsearch separately), which is exact for the interval rather than a sampled percentage. In this campaign it covers **every cell of the main session** (floors, S1, S1m, S1r, S2, S3, S4 — throttling counters included), the topology arm and the paging A/B. S5, S6 and the joins remain client-side only — their harnesses rebuild container topologies mid-run — so reading *those* rows as total system cost is unsupported.
- **Warm cache only, by construction.** No cold-cache arm exists, so nothing here describes a first query after a restart.
- **The constrained-memory and concurrency scenarios run their client inside a Linux container**, while the rest run it natively on macOS. Wall-clock is comparable within each group and only indicative across them, and the peak-memory metric changes with the platform.
- **The ten-million-row scenarios are not reproducible on the Community tier.** They require a licence whose result quota exceeds 10M — S1, S1r, S2, S5 and S6. Everything else is open: **S3, S4 and J2 reproduce with no licence at all**, and the self-service 30-day Pro trial covers S1m and the remaining joins (§9 sets out the full ladder). So what a third party cannot re-run independently is the headline extraction, not the structural result. What is available instead of a re-run is the evidence: **every measured run of every session is published in the repository** under `results/<session>/` — one JSON per run carrying its wall, CPU, memory, wire bytes, host load and memory-pressure reading, alongside the equivalence gate, the ES|QL truncation and join probes, the connect probe, the engine-quiescence records and the sidecar image digest. The dispersion behind every median, the gates that had to pass before a number was kept, and the state of the host while it was measured are all auditable without a licence. That is not reproduction, and it is not nothing — and reproduction of those five cells is a request away: a time-limited Enterprise evaluation licence, with no contractual obligation attached, is requested from your account on the licence portal (§9).
- **The headline gap is mostly shard parallelism, and it is a recent gain rather than an architectural constant.** The main results run on a 6-shard index, where Trino's connector gets six real splits across two workers (verified live) and our client opens six concurrent point-in-time slices. On a single shard the two engines are close — 1.32×, not 3.75× — so a reader whose index is not sharded should size the extraction advantage from the 1-shard column, not the headline. Note what bounds Trino here: the *shard* count, not the node count — additional workers cannot split a single reader.

  This is not left as a caveat: **section 7 publishes the measured sensitivity run** — the same corpus regenerated from the same seed into a 1-shard index, the engines otherwise unchanged — with the split distribution captured live from `system.runtime.tasks` on *both* topologies: 6 splits across two workers on the 6-shard index, 1 split on the 1-shard one. Both engines get faster with shards, very unequally: ours

3.50×, Trino's **1.23×**, so the S1 gap widens from **1.32× to 3.75×**. Peak client memory (±0.3%) does not move; our client CPU does (3.56 s → 2.83 s — section 7 explains the scheduling); the `GROUP BY` still moves 27.1 KB against 1.39 GB. Trino's own largest gain in the whole benchmark appears there too — its aggregation wall-clock improves **5.2×** — and is published as such. **Section 7.1** then attributes our share of it on our own side alone: the same build with concurrent paging disabled takes 35.34 s against 10.48 s, so the feature is worth **3.37×**, with the non-paging controls identical across arms.

The distinction the run establishes: **wall-clock and our client CPU are topology-sensitive; client memory and pushdown are not.** The client is one process consuming one wire format however large the cluster, and the connector pushes no aggregation down at any shard count. That is now measured rather than argued.

- **Trino's spooling protocol was not measured, and the connectorx comparison does not substitute for it.** connectorx parses the *same* row-oriented client protocol, faster, in Rust; the spooling protocol changes the protocol itself — the leg this report identifies as Trino's bottleneck. It is the unmeasured variant most likely to narrow the S1 gap, and it is queued. Until it is measured, Trino has been benchmarked on its fastest *client*, not necessarily its fastest *path*.

B Appendix — Issues found and fixed while benchmarking

Building this benchmark surfaced several issues in SoftClient4ES. All were fixed, and the results in this report were measured on the released `0.3.0` build that contains the fixes.

Extraction performance

- **Deep-paging sort regression (Elasticsearch 8+).** For a query with no `ORDER BY`, the streaming pager injected `_shard_doc` as the primary sort. From Elasticsearch 8 / Lucene 9 onward this defeats the document-id skip optimisation, so each page re-scanned the whole index instead of seeking to the next position — about **60 ms/page instead of 8 ms/page** on the 10M-row index, an effect that grows with index size. The fix uses `_doc` as the sort under a point-in-time context (which Elasticsearch turns into a total order via an automatic tiebreaker), restoring efficient paging while keeping results complete. A multi-shard completeness test guards against regression.
- **Schema probe re-ran bounded statements.** To advertise a result's schema the Flight endpoint executed the statement itself; when the statement already carried a `LIMIT` it was executed in full, so a bounded extraction read every row off Elasticsearch **twice**. The probe now reads at most 100 rows and never more than the statement asked for, and aggregation-shaped statements — where the limit is the bucket count, not a row count — are left untouched.
- **Per-row conversion cost.** The row-to-Arrow path did redundant per-row work (repeated map normalization and re-parsing of each response page). This was reduced to a single parse per page and a single-pass row conversion, which is the main reason the client-CPU figures in S1/S2 are as low as they are.

Result completeness

A family of cases silently returned incomplete results and were corrected:

- `GROUP BY` without `LIMIT` returned only the first ~10 groups (the default aggregation size).
- Window `PARTITION BY` truncated to the first ~10 partitions.
- `SELECT` without `LIMIT` on the non-scroll path returned only the first ~10 rows.
- An explicit `LIMIT` above `index.max_result_window` failed while the same query with no `LIMIT` succeeded.

Each now returns the complete result set, and each is covered by a multi-shard completeness test. These matter for a benchmark because a silently truncated result would otherwise look like a fast, correct run.

Result shape

Extraction rows previously carried Elasticsearch hit-metadata columns (`_id`, `_index`, `_score`) in addition to the selected columns. Rows now contain exactly the columns the SQL projects, so the client receives — and pays for — only the requested data (8 columns for the S1 query).

Environment note

The published sidecar image originally shipped without a logging configuration, which caused it to log at DEBUG and write every Elasticsearch response byte to the container log — a large performance and data-exposure problem. The image now ships a logging configuration that keeps wire-level logging off by default; the benchmark runs the sidecar as shipped.